

Jaku-in: A Cultural Skills Training System for Recording and Reproducing Three-dimensional Body, Eye, and Hand Movements

Sotaro Yokoi
SONY CSL Kyoto
Japan
University of Tokyo
Japan
yokoi@cybert.t.u-tokyo.ac.jp

Natsuki Hamanishi
SONY CSL Kyoto
Japan
hamanishinatsukilab@gmail.com

Kaishi Amitani
SONY CSL Kyoto
Japan
amiumi2828@gmail.com

Jun Rekimoto
SONY CSL Kyoto
Japan
University of Tokyo
Japan
rekimoto@acm.org

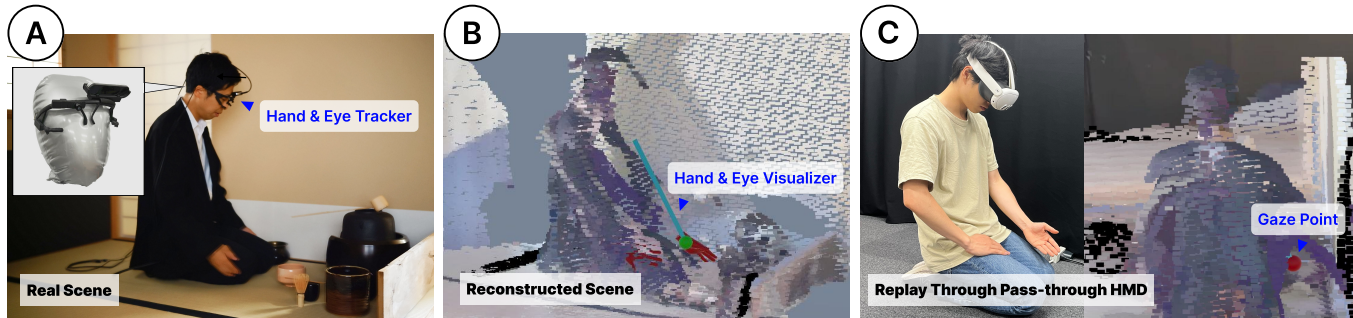


Figure 1: (A) The expert performs the skill with eye and hand trackers in a room where the point cloud can be measured. (B) The reconstructed scene displays the expert's body point cloud, eye, and hand movements. (C) Novices can use the pass-through HMD to play back recordings and observe an expert's body from various viewpoints.

Abstract

In skill acquisition, it is crucial not only to mimic an expert's body movements but also to understand their underlying intentions, which can be achieved by observing their hand movements and gaze. However, it was difficult to understand the difference between one's posture and that of the expert in previous systems that assisted in understanding the intent of the expert. To overcome this limitation, we propose Jaku-in, which displays the expert's hand movements and gazes in a three-dimensional reconstructed space using a point cloud. The system's pass-through functionality of head-mounted displays allows novices to observe and emulate a life-size expert from multiple viewpoints: first-person, third-person, and behind-view. This research initially focuses on the Japanese tea ceremony, a practice where the precision and beauty of physical movements are fundamental. This showcases the effectiveness of 'Jaku-in' in

this domain and inspires potential applications in other intricate skill domains.

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1 Introduction

Effective skill transmission, especially in industries and traditional cultures, hinges on the ability of novices to imitate expert movements accurately. Generally, skills are passed down through video recordings or face-to-face interactions. However, a significant challenge in this process is the effective communication of the expert's intent, which is often subtle and complex. Research indicated that visualizing eye gaze and hand movements can convey these underlying intentions effectively. For example, Matsuda et al. demonstrated that visual cues from an expert's gaze can significantly enhance a novice's understanding of the intentions behind specific hand manipulations in surgical settings [Matsuda et al. 2021]. However, simply watching a video does not allow the user to understand the

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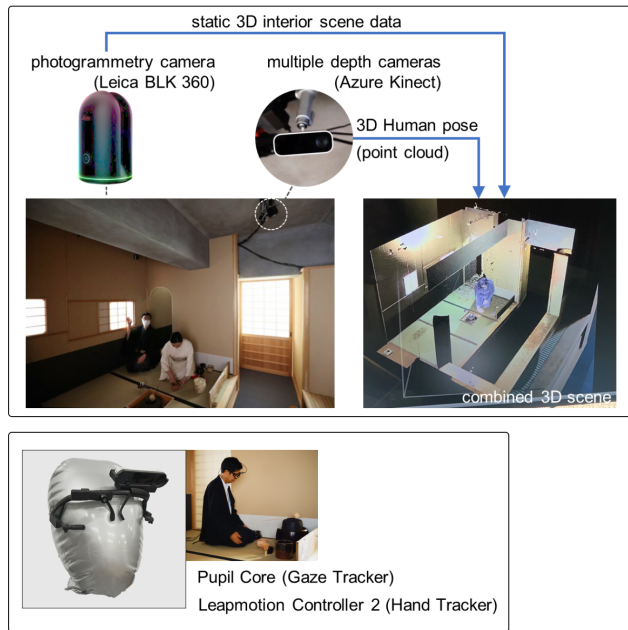


Figure 2: Schematic diagram of the system. There are systems that record body movements using point clouds and systems that record the body movements of skilled workers.

differences in posture and movement between themselves and an expert, which has hindered their skills acquisition.

On the other hand, three-dimensional (3D) reconstruction using point clouds has been proposed to understand users' posture and movements. Komiyama et al., for instance, introduced a system that employs multiple depth cameras and a head-mounted camera, enabling an immersive experience that combines first-person and third-person views for a comprehensive understanding of remote operations through 3D reconstruction [Komiyama et al. 2017]. Despite these advancements, existing 3D reconstruction systems rarely allow observing gaze and hand movements, which is crucial for indicating an expert's intent.

To address this gap, we introduce 'Jaku-in,' a novel system that synchronizes and records the point clouds of an expert's body along with their gaze and hand movements. It is capable of replaying these recordings. This system not only simplifies the archiving of expert movements but also enhances the learning experience by allowing novices to observe these movements from multiple perspectives—first-person, third-person, and behind-view.

This system has been specifically applied to the Japanese tea ceremony. In this art form, the precision and aesthetic of movements are crucial, thus providing a practical and impactful learning tool for preserving cultural heritage.

2 System Design

The system consists mainly of a recording system and a playback system. The system configuration diagram is shown in Figure 2.

2.1 Recording System

The recording system integrates several technologies to capture a comprehensive dataset of the expert's movements. It employs

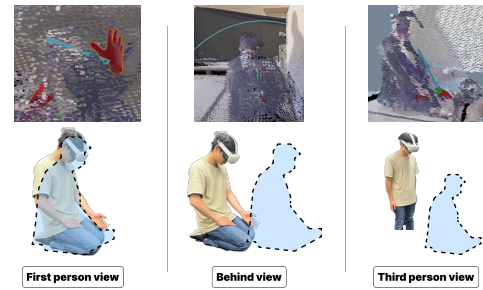


Figure 3: The system offers multiple viewpoints for user interaction: first-person, behind, and third-person perspectives. Each viewpoint is illustrated at the top, with a corresponding overview at the bottom.

four Microsoft Azure Kinect cameras strategically placed around the space to ensure complete coverage without blind spots. Eye movements are captured using Pupil Core, and hand movements are tracked with the Leap Motion Controller 2. Additionally, Visual Simultaneous Localization and Mapping techniques, using an external camera from Pupil Core, are utilized to track the user's head coordinates. The spatial relationships among the world camera, gaze-tracking camera, and Leap Motion Controller 2 were pre-calibrated using markers. Subsequently, this calibration data was utilized to integrate the collected data into a unified coordinate system. Data synchronization and recording are facilitated through custom APIs developed in Unity, ensuring seamless integration of all sensory inputs.

2.2 Replay System

The replay system employs the Meta Quest 3's pass-through feature to superimpose the recorded movements onto the real world, creating a life-sized, immersive learning environment. As depicted in Figure 3, novices can select from multiple viewing angles: a first-person view for direct emulation of movements, a third-person view for understanding spatial context, and a behind view for a complete overview of the expert's body dynamics. To address instances where the expert's rapid eye movements might lead novices to lose track of the gaze direction, we use a visualization technique known as "Halo" [Baudisch and Rosenholtz 2003]. This method represents off-screen gaze points with an arc drawn on the screen, effectively guiding the novice's attention. Moreover, the intricate hand movements of the expert are meticulously overlaid within the user's field of view, allowing for detailed observation and learning.

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